

ANNIE LI

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EDUCATION

Texas A&M University

College Station, TX

Bachelor of Science in Electrical Engineering; Minor in Mathematics, Graphic Design

May 2027

- Craig and Galen Brown Engineering Honors, President's Endowed Scholar
- **Relevant Coursework:** Computer Architecture, Security of Embedded Systems, Digital Integrated Circuit Design, Electronics, GPU Programming, Machine Learning, Fourier Series & Wavelets

EXPERIENCE

Apple

Cupertino, CA

SoC Validation Intern

May 2026 – Aug. 2026

- Engineered **Python** fault extraction pipeline for SoC voltage/temperature sweeps, auto-appending an analysis job with every sweep to triage and cluster **1,200+** DUT logs daily into distinct failure modes
- Authored YAML to isolate both known and novel faults, raising LLM triage accuracy to **89%** on labeled corpus
- Built queryable catalog accumulating **6,000+** faults to surface repeat defects across SoCs, rails, and perfstates

Hewlett Packard Enterprise

Houston, TX

Electrical Hardware Engineering Intern

May 2025 – Aug. 2025

- Prototyped EOL component replacement by wiring in vendor eval board via **Cadence Allegro** net tracing
- Wrote **I2C** scripts to bring up component by sequencing power rails and validating functionality
- Implemented fault detection logic and scan chain-based bus communication on CPLD with **SystemVerilog**

Secure and Trustworthy Hardware Lab

College Station, TX

Undergraduate Research Assistant

Mar. 2024 – Dec. 2025

- Located **2** system-level vulnerabilities on open-source RISC-V processor design using hardware fuzzing framework
- Designed and implemented **5** hardware vulnerabilities into **Verilog** SoC designs for CTF **Hack@DAC 2024**

Texas A&M Health Science Center

Houston, TX

Research Assistant

Jun. 2024 – Aug. 2024

- Developed pattern detection algorithm to automate electrocardiographic data analysis with **MATLAB**

PROJECTS

Tritone | ESP32, C++, FastAPI, WebSocket

- Built **ESP32** assistive listening system to capture speech and deliver live transcription with speaker direction
- Implemented **C++** algorithm for real-time triangulation, detecting a speaker's angle from 3 microphones

Traffic Light Controller | Verilog, Vivado, FPGA

- Designed traffic light FSM that evaluates sensor input and timing conditionals to drive state changes with **Verilog**
- Verified functionality by deploying design onto FPGA using **Xilinx Vivado** with LED output visualization

ACTIVITIES

TAMUhack

College Station, TX

Creative Lead

Nov. 2023 – Present

- Led team of designers in developing visual identity and marketing materials for hackathons with **800+** attendees, resulting in a **17%** year-on-year increase in applications from diverse fields in technology
- Designed **4** accessible and responsive hackathon event websites that draw **20,000+** visitors annually

Institute of Electrical and Electronics Engineers (IEEE)

College Station, TX

Public Relations Chair

Aug. 2024 – May 2026

- Directed creative strategy and external communication for career and social events throughout the year

SKILLS

Programming: C++, C, Python, SystemVerilog, MATLAB, HTML, CSS

Softwares: Cadence Allegro, Vivado, SolidWorks